VB037IF3000 – Worked Example of Every Sheet in fert_report.xlsx

This document follows **one FERT, VB037IF3000** (plan week 2026-44, priority 1, shop SH-3, plant AC02) through the whole workflow: from the three SAP tables, through the stock assignment, into every sheet of `fert_report.xlsx`.

Source of the numbers. All figures marked *report* are read from your `fert_report.xlsx` and the supplied ZPLAN / ZMB52 / ZBOMEXPLODE files. Section 10 compares them with what the program itself produces, and lists the differences honestly.

Terminology. *Sheet* = Excel sheet name. *Column* = Excel column heading, written exactly as in the report (e.g. `Req.Qty as per BOM`). Table / field names (`zplan`, `Plan_week`, …) refer to your MySQL tables.

1. The plan line (ZPLAN → sheet PLAN)

Table `zplan` holds one row for this FERT (Excel row 372 of ZPLAN.xls):

`zplan` field	Value	Meaning
Plant	AC02	plant
Material	VB037IF3000	the FERT
Plan_week / Plan_month	2026-44 / 2026.10	week and month of the plan
Priority	1	order of allocation inside the week (all shops share one numbering)
Plan_Qty	5	motors to be built
Plan_Value	55,764.90	5 × 11,152.98 (MM_Price)
Original_FG_Stock / Assigned_FG_Qty	0 / 0	no finished-goods stock exists for this FERT
FG_Stock_after_Assignment	0	
Balance_Plan_Qty_to_be_Produce	5	still to be produced
Shop / Fert_or_Spare_Plan / Product	SH-3 / FERT / ACM	

The same line appears in sheet **PLAN**, FERT block, as row Srl. 191:

Column in PLAN	Value	Where it comes from
`Srl.`	191	running number inside the FERT block
`PLANT`	AC02	`Plant`
`Product` / `Product Desc`	ACM / AC MOTORS	`Product` / lookup file `product_desc.csv`
`FERT` / `FERT Desc`	VB037IF3000 / 2.2KC4,BRC100V,415V,50Hz,55/411,IE3	`Material` / `Material_Description`
`Plan Week` / `Plan Month`	2026-44 / 2026.10	`Plan_week` / `Plan_month`
`Del Date`	26.10.2026	`Requested_deliv_date`
`Priority`	0001	`Priority` written with 4 digits
`Plan Qty` / `Plan Val`	5.00 / 55764.90	`Plan_Qty` / `Plan_Value`
`Assigned Qty` / `Assigned Value`	0.00 / 0.00	`Assigned_FG_Qty` / `FG_Assigned_Value`
`Remark`	blank	not used

PLAN is sorted by `Plan Week`, then `Priority`; the FERT block comes first, then a second block headed `SPARE` / `SPARE Desc` for spare lines.

2. Finished-goods stage and the FG ratio

FG stock is allocated first: `Assigned_FG_Qty = MIN(Plan_Qty, stock of the material)`. For VB037IF3000 the stock is 0, so

```
FG ratio = Balance_Plan_Qty / Plan_Qty = 5 / 5 = 1
```

The ratio scales the whole BOM. Example from the same data: spare A0222057810501 (week 2026-44, priority 37) has plan 5 and FG stock 4 → assigned 4, balance 1, ratio 0.2, so only 20 % of its BOM is required.

3. The BOM structure of VB037IF3000 (ZBOMEXPLODE)

The explosion has 124 rows (Level 1 to 8). Rows are in explosion order (`Running_Serial_number`); the parent of a row is the nearest preceding row one level higher. Key codes in the BOM:

- **E** = in-house item, **F** = brought-out (purchased) item.
- **Phantom** (`Phantom_item` = x, special procurement 50) = grouping assembly with no stock of its own; its components belong to the grandparent.
- **Special procurement 30** = sub-contract item; its components are still bought and shown.
- **Text-only line** = BOM line without a material code (instructions such as "DRAIN HOLE").

Condensed tree (components counted as they appear in the Brought Out sheet):

```
VB037IF3000 (FERT, plan 5)
├─ M322120155CBPK  phantom (packing)            7 brought-out items
└─ MVB037IF3000FA  Final assembly (E)           4 items
   ├─ M00000100280FP  phantom (paint, notes)    4 items + 2 text lines
   └─ MVB037IF3000GA  General assembly (E)      9 items
      ├─ CF0224S30YDF5020  Stator assembly (E)  2 items + sub-levels
(winding, insulation kit)
      ├─ H03294140AY1DR1I  Rotor assembly (E)   2 items + sub-levels (shaft,
rotor stack)
      ├─ MA3P01000001TX  phantom                6 items
      ├─ MBRC100L000HKA  phantom (bearings, fixings)  20 items + 11 text
lines
      └─ S0M50U20003GKT phantom                  4 items
```

In total the report shows **87 brought-out rows** and **22 in-house rows** for this FERT (9 in-house assemblies + 13 text lines).

4. How the required quantity is calculated

```
Req.Qty as per BOM = Bom_Qty × SA_Qty × Plan_Qty        (column
Total_Org_BomQty)
R (needed after FG and sub-assembly stock) = Req.Qty as per BOM × FG ratio ×
coverage factor of the parent
```

Example A – no reduction. Component A0322205170000 (cover NDE) under MVB037IF3000GA: Bom_Qty 1 × SA_Qty 1 × Plan 5 = **5**. FG ratio 1, parent not covered by stock → R = 5.

Example B – reduction by sub-assembly stock. The brought-out sub-assembly **H0329484P140YR** (rotor stack, special procurement 30) needs 5. Plant stock is 3 and 2 are assigned to this line (the other stock went to earlier lines), so 3 are short:

```
coverage factor of its children = (5 − 2) / 5 = 0.6
```

Its child **A40CR50C400000** (CRNGO steel, group CRN) has `Req.Qty as per BOM` = 1.85 × 4.2 × 5 = **38.85**. Because only 60 % of the rotor stack must still be built:

```
R = 38.85 × 0.6 = 23.31          → report: Assigned Qty 23.310, Shortage Qty
0.000
SA Required Qty (parent need) = 21 × 0.6 = 12.600   → report column `SA
Required Qty` = 12.600
```

The `Req.Qty as per BOM` column still shows the original 38.850; the reduced figure shows up in Assigned/Shortage.

5. How stock is assigned – the ledger of three components

Stock is a **plant-wide pool per material**. Lines are served in order **Plan Week, then Priority** (all shops together). VB037IF3000 has priority 1 but is in week 2026-44, so it is served **after** every line of earlier weeks. For each line:

```
Assigned = MIN(R, stock still left in the pool)       Shortage = R − Assigned
```

5.1 A1422098070100 – bearing cap, fully covered

Stock (ZMB52) 42, group BC, purchasing group 302. The report allocates it over 10 plan lines; VB037IF3000 takes 5 (needed 5) → shortage 0. Total assigned over all lines = 23 = ZMB52 `Assigned_Stock_Qty` (23); left 19.

5.2 A0322205170000 – cover NDE, partly covered

Stock 7. Lines in the order they are served (all rows from the Brought Out sheet):

Order	Plan Week	Priority	FERT	Req.Qty as per BOM	Assigned Qty	Stock left after
1	2026-38	29	RC037AF3000	4	3	4
2	2026-43	11	SH266080	1	1	3
3	2026-43	52	RB037AF3100	1	1	2
4	**2026-44**	**1**	**VB037IF3000**	**5**	**2**	**0**

(Other lines of this component in weeks 38/39 need 1 each but have R = 0 because their FERT came from FG stock.) VB037IF3000 needs 5, only 2 remain → **Assigned 2, Shortage 3**. Sum assigned = 7 = ZMB52 `Assigned_Stock_Qty` (7).

5.3 A0322107140000 – flange, nothing left

Stock 2, both already assigned to earlier lines (ZMB52 shows Assigned 2 / Balance 0). VB037IF3000 needs 5 → **Assigned 0, Shortage 5**.

5.4 What the report shows for these rows

Component (sheet Brought Out)	Req.Qty BOM	Req.Qty MRP	Assigned Qty	Shortage Qty	Price	Latest Price	Price Source
A1422098070100	5.000	0.000	5.000	0.000	14.60	= price	STOCK_VAL
A0322205170000	5.000	5.000	2.000	3.000	323.44	325.00	STOCK_VAL
A0322107140000	5.000	5.000	0.000	5.000	526.22	536.00	STOCK_VAL

Rule for the MRP column: `Req.Qty as per MRP` = R when Shortage > 0, otherwise 0. That is why a fully covered item shows 0 and a short item shows its **full** requirement (5, not 3).

5.5 Other situations visible in this FERT

- **No stock record at all** – D1322222150002 (group ZPV): not in ZMB52 → Assigned 0, Shortage 5. Price 6.50 comes from the material master (`Price Source` = MM_PRICE), so `NO InfoRecord` = **NO-PRICE-LINK**.
- **Stock exists but all consumed** – see 5.3.
- **Partial** – see 5.2 and the rotor stack in section 4.
- **Dependency matrix** – rows under MVB037IF3000GA carry two codes (`Dependency_Matrix_Code_1/2` = CF0224S30YDF5020 and H03294140AY1DR1I, the stator and rotor assemblies). In this FERT the report assigns bearing cap A1422098070100 (5 of 5) even though the rotor assembly itself got no stock, so **your report did not limit components by the dependency codes here**. The program therefore has the dependency rule as a tick-box that is **off by default**.
- **Sub-contract parent (special procurement 30)** – C0009901230000 (copper rod) is reported under parent U2C00050000800 (SA Required 5.125, Assigned 5.125 from stock 11,961) and under U2C00050001000 (SA Required 0, Assigned 0).

6. Column dictionary – sheet "STOCK ALLOCATION-Brought Out" (35 columns)

Example row: component **A0322205170000** of VB037IF3000.

#	Column	Example	Source / calculation
1	`Srl.`	running no.	1, 2, 3 …
2	`PLANT`	AC02	`plant`
3	`Product`	ACM	`Product`
4	`FERT`	VB037IF3000	`Fert`
5	`FERT Desc`	2.2KC4,BRC100V,415V…	`Fert_Description`
6	`Plan Week`	2026-44	`Plan_week`
7	`Plan Month`	2026.10	`Plan_month`
8	`Del Date`	26.10.2026	`Requested_deliv.date`
9	`Priority`	0001	`Priority`, 4 digits
10	`Plan Qty`	5.00	`Plan_Qty`
11	`Parent Code`	MVB037IF3000GA	`Parent_Code` (immediate parent in the tree)
12	`Parent Desc`	GENERAL ASSEMBLY - BRC100 - VB037IF3000	`Parent_Material_Desc`
13	`SA Required Qty`	5.000	need of the parent: its `Total_Bom_Qty` × FG ratio × coverage (first level: `SA_Qty` × `Plan_Qty`)
14	`Component`	A0322205170000	`Component`
15	`Component Desc`	COVER NDE,RC100,32220517,CI,M/C,OIL-SEAL	`Material_Description`
16	`Material Group`	CNM	`Material_Group`
17	`Mat.Grp.Desc`	COVER NDE-CI M/C	lookup `mat_group_desc.csv`
18	`UOM`	PC	`Base_Unit_of_Measure`

#	Column	Example	Source / calculation
19	`Price`	323.44	`Unit_price`
20	`Pur.Group`	304	`Purchasing_Group`
21	`Pur.Name`	Chandrashekar S	`Description_p_group` in ZMB52 for that group
22	`Proc.Typ`	F	`Comp_Code_Proc_Type`
23	`Spl.Proc.Typ`	blank	`Comp_Code_Spl_Proc_Type`
24	`ABC Class`	A	`ABC_Indicator`
25	`Req.Qty as per BOM`	5.000	`Total_Org_BomQty`
26	`Req.Val as per BOM`	1,617.20	`Total_Org_BomVal` (= 5 × 323.44)
27	`Req.Qty as per MRP`	5.000	R if shortage > 0, else 0
28	`Req.Val as per MRP`	1,617.20	`Req.Qty as per MRP` × `Price`
29	`Assigned Qty`	2.000	stock assigned (section 5)
30	`Assigned Value`	646.88	`Assigned Qty` × stock price (ZMB52 `MM_Price`; `Price` if that is 0) = 2 × 323.44
31	`Shortage Qty`	3.000	R − Assigned
32	`Shortage Value`	975.00	`Shortage Qty` × `Latest Price` (3 × 325.00)
33	`Price Source`	STOCK_VAL	`Price_source` (STOCK_VAL / PUR_INFO / MM_PRICE)
34	`Latest Price`	325.00	`Latest_Inforecord_Price`
35	`NO InfoRecord`	blank	NO-PRICE-LINK when Price Source = MM_PRICE; MISSING-VIEW when Price Source is blank

Two valuation rules worth remembering, both checked on your whole report:

- **Shortage Value uses `Latest Price`** (100 % of 12,258 shortage rows). A0322107140000: 5 × 536 = 2,680.00, not 5 × 526.22.
- **Assigned Value uses the stock price.** H0329484P140YR: `Price` 50.40 but stock price 762.96 → 2 × 762.96 = **1,525.92**.

Rows are sorted by `Plan Week`, `Priority`, parent level, `Parent Code`, `Component`. Which rows appear: brought-out (`Proc.Typ` F) items that are not phantom, not below a brought-out parent without special procurement 30, filtered by Shop and Purchase group.

7. Totals for VB037IF3000 in the Brought Out sheet (87 rows, in rupees)

Column	Sum
`Req.Val as per BOM`	52,062.74
`Req.Val as per MRP`	12,106.90
`Assigned Value`	39,186.18
`Shortage Value`	11,354.33

61 of the 87 rows have `Shortage Qty` > 0, and exactly these 61 appear in the next sheet.

8. All nine sheets with VB037IF3000

8.1 SHORTAGE LIST

Contains the Brought Out rows with `Shortage Qty` > 0 (61 rows for this FERT). Column names are the same except `Required Qty` / `Required Val` (= `Req.Qty / Req.Val as per BOM`), and there is no MRP or Latest Price column. Example: A0322107140000 – Required 5.000 / 2,631.10, Assigned 0, Shortage 5.000 / 2,680.00.

8.2 STOCK ALLOCATION- In house (33 columns)

Same layout as Brought Out up to `Shortage Value`, then `Routing Group`. It holds items where `Proc.Typ` is not F: the 9 assemblies and the 13 text lines of this FERT. Examples:

Parent Code	Component	Req.Qty as per BOM	Req.Val as per BOM	Assigned	Shortage	Routing Group
VB037IF3000	MVB037IF3000FA (final assembly)	5.000	54,151.40	0	5	20200214
MVB037IF3000GA	CF0224S30YDF5020 (stator assembly)	5.000	35,898.00	0	5	20200180
M00000100280FP	(blank) "USE SHADE EX-280 OF IS-5 TO MAIN MOTOR"	5.000	0.00	0	0	blank

Assemblies show a value that includes everything below them, so the in-house values must not be added up with the brought-out values. No assembly of this FERT has stock, so Assigned is 0 and Shortage 5 on each. `Routing Group` comes from SAP routing data that is not in the three tables; the program leaves it blank.

8.3 USABLE & UNUSABLE STOCK

One row per ZMB52 row. For A1422098070100: `Total Qty` 42.000, `Total Val` 613.21, `Usable Qty` 23.000 (sum of Assigned Qty over all plan lines, including the 5 for VB037IF3000), `Usable Value` 335.81, `Unusable Qty` 19.000, `Unusable Value` 277.40. For A0322107140000: Total 2, Usable 2, Unusable 0. Formula: `Unusable = Total − Usable`; `Usable` = assigned in Brought Out + In house + PLAN (`Assigned Qty`).

8.4 MOQ VAL

Brought-out items that have a minimum order quantity. A1422098070100 appears with `MOQ Qty` 100, `Unit Price` 14.60, `Total Stock Qty` 42.000, `Shortage Qty as per BOM` 0, `As Per MRP Qty` 32.000 (value 467.20), `Addtional MOQ Qty` 32.000 (value 467.20). The MRP quantity and the extra MOQ quantity come from SAP's lot-size logic (MOQ and reorder point). The program only **approximates** that figure, so this sheet is the least certain one.

8.5 NO PRICE LINK

Should list brought-out items bought without an info record. D1322222150002 is flagged `NO-PRICE-LINK` in the Brought Out sheet (shortage 5, price 6.50 from the material master) but **does not appear** in your sample NO PRICE LINK sheet; the inclusion rule of that sheet could not be derived from the data.

8.6 MISSING VIEW

Items whose `Price Source` is blank (no MRP or purchasing view). None of the components of VB037IF3000 qualifies.

8.7 Summary (values in lakhs, month column OCT-2026 = Plan Month 2026.10)

This FERT contributes: `Sales Plan Qty` 5, `Sales Plan Value` 0.56 lakh; `Brought out Materials-as per BOM` 0.52; `as per MRP` 0.12; `Procured - Value of Stock used for Current Plan` 0.39; `Total Shortage Value` 0.11 (52,062.74 / 12,106.90 / 39,186.18 / 11,354.33 ÷ 100,000).

9. Typical scenarios – where to find them in this FERT

Scenario	Example row	Result
Fully assigned	A1422098070100	Assigned 5, Shortage 0, MRP 0
Partly assigned	A0322205170000	Assigned 2, Shortage 3, MRP 5
Stock all used by earlier lines	A0322107140000	Assigned 0, Shortage 5
No stock record	D1322222150002	Assigned 0, NO-PRICE-LINK
Reduced by sub-assembly stock	A40CR50C400000	38.85 → Assigned 23.31
Value uses stock price	H0329484P140YR	2 × 762.96 = 1,525.92
Same component twice in one parent	U2B10007050410 under MBRC100L000HKA	two separate rows of 10
Phantom parent	MBRC100L000HKA	components shown under the phantom, `SA Required Qty` 5
Text-only line	"DRAIN HOLE" under MBRC100L000HKA	In-house sheet, values 0
In-house assembly without stock	MVB037IF3000GA	Assigned 0, Shortage 5
MRP larger than BOM	U2IM009907840Z (4 rows)	`Req.Qty as per BOM` 0.175 but `Req.Qty as per MRP` 0.240, because SAP bases it on the parent's need (240)

10. Report versus program output for this FERT

The program applies the same rules, but your ZBOMEXPLODE file contains only week 2026-44, so the stock consumed by earlier weeks (section 5.2 shows this matters) is not consumed first. Result for VB037IF3000 (87 rows):

Sum	Your report	Program from the supplied files
Req.Val as per BOM	52,062.74	52,062.74
Req.Val as per MRP	12,106.90	6,229.12
Assigned Value	39,186.18	40,879.70
Shortage Value	11,354.33	5,206.01

64 of 87 rows are identical; 23 differ, almost all because the program sees more stock than was left at week 44 (for example A0322205170000: program 5, report 2). With the full BOM export (all weeks) the order of consumption is reproduced and these rows should agree. Other known differences: the 4 `U2IM009907840Z` rows (MRP basis above) and `U2C20007000500` / `A40CR50C400000` (coverage of sub-assembly stock is approximated). Please compare once with the full ZBOMEXPLODE before relying on the figures.